



FT31 – Fluidised Bed Dryer

The FT31 is a simple, compact fluidised bed dryer, conveniently portable and easy to operate.

- Small batch sizes for minimised wastage
- Small footprint for fitting onto a benchtop
- Rapid start up for faster results

PRODUCT CODE: FT31

Categories: Armfield Dairy, Armfield Educational Food Technology, Armfield Flavours And Fragrances, Armfield Nutraceuticals, Armfield Pharmaceuticals, Educational, FT Series - Educational Food Technology, FT31 Laboratory Fluid Bed Dryer, IFT Industrial Food Technology, IFT Series - Industrial Food Technology

Tags: Agglomerating Materials , Agglomeration , Centrifugal Fan , Classifier , Drying , Fluidised Bed Dryer , FT31 , Heat And Mass Balance , Humidity Probe , Multitub , Thermal Efficiency , Thyristor

DESCRIPTION

The dryer is of simple, compact design, conveniently portable and easy to operate, the only requirement being a mains power supply.

Air is drawn through a mesh filter in the base of the cabinet and blown by a centrifugal fan over a 2kW finned electrical heater and through a stainless steel filter gauze. The air is then delivered to the distributor gauze at the base of the dryer body which supports the bed and distributes the air uniformly.

The air blower is controlled by a thyristor circuit to give a smooth variation over a wide range of motor speeds, enabling efficient fluidisation to be achieved for a variety of materials and giving fine control of the drying temperature. Readings are selected and displayed using a digital meter.

The unit can be manually operated or interval timing can be carried out with the timer unit, which gives a 0-10 minute timing range and an alarm facility to notify completion of the operation.

The tube unit locks into position on the cabinet top by a simple bayonet fitting and the base of the tube is removable to allow replacement of the distributor gauze. A filter bag is employed to retain any stray particles of the sample being fluidised, allowing the passage of the exit gases.

TECHNICAL SPECIFICATIONS

Power Consumption: 3kW

Temperature Range: 20-200°C ($\pm 1^{\circ}\text{C}$)

Timer Range: 0-10 minutes

Max Sample Weight: 5kg

FEATURES & BENEFITS

- High rates of heat and mass transfer
- Drying times and therefore residence times range from a few seconds to a few minutes – often less than 15 minutes are required for complete drying
- Materials with moisture contents from a few percent to over 80% may be dried
- The drying air temperature range is ambient to 200°C which covers the majority of drying operations
- A wide range of materials can be processed including organic and inorganic compounds, pharmaceutical chemicals, foods and fuels, minerals and agglomerating materials
- In addition to drying, other processes may be investigated, eg mixing and blending of solids, size reduction, agglomeration and granulation

Demonstration Capabilities

- Simple drying of a material to give moisture content and the drying time (or residence time) required
- Determination of drying curves to assess the feasibility of fluidised bed drying of a material on an industrial scale. Drying curves are irrelevant to the mechanism of drying – they may be used as a basis for heat and mass balance, thermal efficiency of drying and dryer design
- Calculation of heat transfer coefficients for different conditions – important in dryer design and comparison of fluidised beds with other drying methods

Applications

- Powders
- Granular products
- Particulates
- Cereals and grains
- Pigments and dyes
- Biomass
- Metal powders
- Catalysts, resins, and polymers

Requirements

ELECTRICAL SUPPLY: SINGLE PHASE (SEE ORDERING CODES)

FT31-A: 220-240V / 1ph / 50Hz

FT31-B: 120V / 1ph / 60Hz

FT31-G: 220-240V / 1ph / 60Hz

Shipping Specifications



PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 0.6m³

Gross weight: 80kg

Dimensions

Length: 0.26m

Width: 0.32m

Height: 0.465m

Order Codes

FT31-A

FT31-B

FT31-G

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